Research Philosophy — Reliable AI and Intelligent Sensing for Connected Engineering Systems

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Research philosophy and established contributions

I view engineering research as the disciplined process of building generalizable methods that remain useful when laboratory assumptions no longer hold. Strong scholarship should make an original technical contribution, expose its assumptions, assess failure modes, and generate artifacts that others can reproduce or extend. At an institution centered on experiential education, it should also create meaningful opportunities for student participation and connections with industry practice.

My PhD in Electrical and Computer Engineering from the University of Oklahoma (2025) developed AI/ML approaches for wireless networks under limited data, domain variation, and changing operating conditions. My scholarship has addressed domain-informed generative augmentation for radio-propagation modeling, data-driven digital twins, outage detection and self-healing, and learning-based network optimization. Related work examines smartwatch inertial signals for activity and medication-adherence recognition. Peer-reviewed outputs include work in Computer Communications, Sensors, IEEE Network, IEEE PIMRC, and IEEE VTC. My research and industrial appointments at AT&T, AT&T Labs, Futurewei, and Impac Exploration Services have provided firsthand experience with imperfect operational measurements, heterogeneous data, the need for explainable diagnostics, and software workflows that must remain reliable outside controlled experiments.

These results form a methodological foundation for a Kettering research program in reliable AI and intelligent sensing for networked and cyber-physical engineering systems. The proposed program would stay grounded in areas where I have published or built systems, while opening interdisciplinary collaborations in smart mobility, industrial monitoring, and autonomous technologies through complementary faculty expertise.

Direction 1 — Data-efficient, trustworthy learning for sensing and communication

Real engineering systems often have abundant raw measurements but few trustworthy labels; their sensors, environments, and usage patterns also change over time. My first direction will investigate methods that use limited measurements more effectively without hiding the risks of synthetic or transferred data. Building on my work in domain-informed generative models, transfer learning, and robustness evaluation, I will study data-efficient adaptation, distribution-shift detection, uncertainty calibration, and principled synthetic-data validation.

Research testbeds would begin with wireless-network and sensor datasets where ground truth and domain conditions can be controlled. Models would be compared not only by average predictive accuracy but also by cross-environment performance, calibration, false-alarm burden, resource cost, and sensitivity to missing measurements. A possible interdisciplinary application is condition monitoring of connected equipment, where reliable inference matters more than impressive performance on a single laboratory dataset. Expansion into machine, battery, or robotics instrumentation would be pursued jointly with subject-matter experts rather than presumed as an existing experimental specialty.

Direction 2 — Adaptive and resilient networked systems

The second direction concerns closed-loop decision making under imperfect observations. Communications and sensing are critical to intelligent vehicles, distributed automation, and industrial cyber-physical systems. Building on my previous research in network self-healing and reinforcement-learning-based optimization, I will study anomaly detection, adaptive resource allocation, safe decision policies, and reliable control-oriented analytics for connected systems.

Early studies would use a reproducible Python-based simulation environment to vary channel quality, traffic, topology, sensing delays, and disturbances. The goal is to understand when a learning-based policy provides a measurable benefit over interpretable rules or conventional optimization, and under what conditions it becomes unreliable. Research questions

include: How should an adaptive system respond when its input distribution changes? When should it defer to a conservative baseline? How can we evaluate resilience, service continuity, and fairness alongside prediction accuracy?

Interdisciplinary collaborations could connect these methods to intelligent mobility and robotic platforms through wireless connectivity, distributed sensing, or decision-support modules. I would keep the role of AI and communication methods explicit and integrate hardware-specific expertise through faculty partnerships.

Direction 3 — Applied AI platforms and student-centered engineering research

A third direction will investigate practical experimental platforms that connect computation, instrumentation, and reproducibility. My work in wireless simulation, network logs, cloud data engineering, and wearable time-series modeling suggests an accessible starting point for undergraduate research: well-scoped datasets, instrumented experiments, interpretable baselines, and documented computational pipelines.

Examples include student-built signal-analysis toolkits, anomaly-detection demonstrators, sensor-fusion studies using appropriately authorized datasets, and industry-inspired network-health monitoring exercises. Each project would have explicit engineering questions, measurable outcomes, appropriate ethics and data protections, and a documented path from exploratory results to verification. Where actual laboratory hardware is available, I would work with experienced faculty and staff to ensure safe, curriculum-aligned use and realistic maintenance demands.

These platforms can serve both research and teaching. Students can help design experiments, test robustness hypotheses, and present results to technical and nontechnical audiences. This aligns with Kettering's teacher-scholar expectations: research should not be isolated from the learning environment, yet a rigorous methodological contribution must remain central.

Research development, partnerships, and funding

Year 1: Establish two reproducible research streams, one on reliable learning under domain shift and one on adaptive connected systems; recruit interested undergraduate students and identify colleagues and co-op partners with complementary instrumentation or application expertise. Develop one benchmark or methods paper and prepare a focused collaborative external proposal after reviewing sponsor eligibility and institutional priorities.

Years 2–3: Extend methodological work into multimodal sensing and resilient networked systems; develop shared experiments and publish peer-reviewed journal or conference articles. Seek external support through eligible federal programs, appropriate foundations, and industry partnerships, potentially drawing on NSF and mission-relevant applied research solicitations when the objectives align. Student roles would span experimental setup, coding, statistical evaluation, documentation, and, for advanced students, original algorithm design.

Years 4–5: Consolidate a sustained research program supported by reproducible tools, interdisciplinary applications, graduate and undergraduate supervision, and competitive collaborative funding. Evaluate progress through rigorous publications, student development, research reuse, and the actual reliability and utility of demonstrated systems, not merely grant count or model size.

I have participated in research and proposal-development activities and received graduate research support; I do not claim to have previously held a sponsored grant as principal investigator. My funding approach is therefore to build a credible independent program through focused, well-evidenced proposals and partnerships, expanding the scope as results and collaborations mature.

Scholarship, mentoring, and service

My mentoring model combines high expectations with well-defined scope. Students would begin by reproducing a result, explaining its assumptions, documenting code and data, and testing a controlled variation. More advanced work would require an independent research question, defensible baseline comparison, uncertainty analysis, and clear authorship practices. I would coordinate research tasks with Kettering's co-op and academic schedules, allowing work to be modular and maintainable when students alternate between campus and professional placements.

I also regard responsible engineering as inseparable from research quality. Data provenance, reproducibility, transparent failure reporting, ethical AI use, and respect for project partners would be embedded from the outset. I would contribute to departmental scholarship, student research experiences, curriculum-linked demonstrations, and collaborative service in ways that support Kettering's values of respect, integrity, creativity, collaboration, and excellence.

My long-term objective is a practical yet scientifically rigorous research program in reliable learning, sensing, and networked systems, one that advances AI methodology while equipping students to evaluate and improve the technologies they encounter in industry.